

# TwinGaze

Find hidden cameras with the phone in your pocket. On the phone, offline, in your language.

## TEAM

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## WHAT THIS IS

Presenter script and cue points for the pitch · app version 3.2.3 · 27 Sept 2026

### How to use this document

**SAY** is the text to read out loud. Short sentences, written to be spoken. Lines marked

*optional* can be skipped if you are short on time.

**KEY POINTS** are the small points the judges must remember. If you lose your place, just say these.

|                                                               |          |
|---------------------------------------------------------------|----------|
| 1 · Problem statement: the example (India)                    | ~1 min   |
| 2 · The problem, and what we solve (scattered apps → one app) | ~1 min   |
| 3 · Our solution: TwinGaze · 3b · every feature               | ~2 min   |
| 4 · Tech stack · 4b · architecture diagram                    | ~1.2 min |
| 5 · AI models · 5b · how the model was made                   | ~1.5 min |
| 6 · Technical details · 6b–6c · flow charts                   | ~2.5 min |
| 7 · Live demo (cue sheet) + closing line                      | ~2 min   |
| 8 · Honest limits and likely judge questions                  | if asked |

The example: what we are dealing with. Start with a person, not with technology.

## SAY

Meet **Ananya**. She is 21. She has just moved from Warangal to Hyderabad for her first job, and she has taken a room in a women's **PG in Ameerpet**.

Her room has a wall clock, a phone charger by the bed, a smoke detector on the ceiling, and Wi-Fi run by the owner. **Any one of them could hide a camera, or a small audio recorder**. These devices cost a few hundred rupees online, and they are small enough to fit inside a charger.

This is not just a story. In 2015, Union Minister **Smriti Irani** found a camera pointed at the trial room of a clothing store in **Goa**. News of cameras found in PG hostels, hotel rooms, trial rooms and washrooms keeps coming from cities across India.

Ananya has no way to check her room. She can only look around. And research shows that people searching by eye find **fewer than half** of hidden cameras.

## KEY POINTS

- **Example:** Ananya, 21, first job in Hyderabad, PG room in Ameerpet
- **What can hide in one room:** a camera · an audio recorder · a tracker · Wi-Fi the owner controls
- **Hiding places:** wall clock, charger, smoke detector, photo frame, bulb holder
- **Cheap and tiny:** a few hundred rupees online
- **Real case:** 2015, Goa: a camera facing a store's trial room, found by Smriti Irani
- **By eye, people find only 46%** of hidden cameras (LAPD study, NUS, ACM SenSys 2021)

Today the tools are scattered across many apps, and none of them can be trusted alone.

### SAY

So what is the problem? **Hidden cameras and recorders are cheap, tiny and hard to see, and ordinary people have no reliable way to check a room.**

Today the tools are **scattered**. There are many separate apps: one that reads the **magnetic field**, another for infrared lights, a Wi-Fi scanner, a Bluetooth tracker finder, an antivirus for spyware, and yet another app for SOS or a fake call. Each does one small thing. Many camera-finder apps only read the magnetic sensor, so they beep near every charger and speaker, and **people stop trusting them**. None of them guides you, none keeps evidence, and most speak only English.

What we solve: **one app** that checks the room, the network and your own phone; that **combines the signals so they confirm each other**; that guides you by voice in your language; and that keeps evidence and gets you help. All offline, on the phone.

| TODAY: A SEPARATE APP FOR EACH     | THE PROBLEM WITH IT                        | IN TWINGAZE                                                                                                                          |
|------------------------------------|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <b>Magnetic-field detector</b>     | Beeps near any electronics: false alarms   | One of four signals; finds hidden electronics up close (cameras, audio recorders, trackers); never calls a camera on magnetism alone |
| <b>IR / camera-finder</b>          | You stare at the screen and judge yourself | Night vision + lens glint, tested automatically (flash, sideways, surface)                                                           |
| <b>Wi-Fi / network scanner</b>     | Technical lists, no meaning                | Camera hotspots, cameras on the network, "is this Wi-Fi private?"                                                                    |
| <b>Bluetooth tracker finder</b>    | Only one kind of device                    | AirTag, Tile, SmartTag trackers and Bluetooth cameras                                                                                |
| <b>Antivirus / spyware scanner</b> | Doesn't say who is watching                | Phone check: stalkerware, tracker SDKs, apps using the camera or mic                                                                 |
| <b>SOS / fake-call app</b>         | Yet another app to install                 | SOS (power button x3), fake call, siren, share location                                                                              |
| <b>(nothing)</b>                   | No guidance, no proof                      | AI shape recognition, voice guide in 3 languages, evidence + complaint                                                               |

### KEY POINTS

- **Problem:** cheap, tiny cameras and recorders; people can't check a room
- **Today:** scattered single-purpose apps; many are magnetic-only → false alarms → no trust
- **We solve it with one app:** room + network + your phone + personal safety · signals confirm each other · your language · offline

One app that replaces all of them: find, confirm, guide, prove, get safe.

## SAY

Our solution is **TwinGaze: one app that replaces all of those**. It turns an ordinary Android phone into a hidden-camera detector. **Everything runs on the phone. Nothing is uploaded, and it works without internet.**

TwinGaze looks for a camera in **four independent ways** at the same time. **One**, it recognises cameras by their **shape**, with an AI model we trained ourselves. That is how it finds the CCTV domes in this hall. **Two**, it catches the **glint** of a hidden lens: a lens throws the phone's flash straight back, like a cat's eye at night. **Three**, the **magnetometer**: powered electronics disturb the magnetic field, so the phone beeps faster as you get closer. That also finds other hidden electronics up close, like an **audio recorder** or a **GPS tracker**. **Four**, in a dark room it sees the **infrared lights** of night-vision cameras.

It also scans **Wi-Fi, Bluetooth and the local network** for cameras and trackers, and checks **your own phone** for spyware and for apps secretly using your microphone or camera. A built-in **agent guides you by voice** in English, Hindi and Telugu. When a camera is found, it saves the photo with a **tamper-proof fingerprint** and writes a ready complaint under the Bharatiya Nyaya Sanhita.

And for your safety: an **SOS** you trigger by pressing the power button three times, even when the phone is locked; a **fake call**, where your phone rings so you have a reason to leave; a **siren**; and one tap to **share your location**.

### Shape

Our AI model boxes CCTV, dome and bullet cameras (and phones)

### Glint

Flash + camera: a lens reflects the flash back

### Magnetic

Cameras, recorders, trackers: beeps faster up close

### Night vision

Infrared LEDs show up as purple dots in the dark

## KEY POINTS

- **One app instead of many · on-device · offline · private:** nothing is uploaded
- **Finds:** hidden cameras (4 ways) · hidden electronics like audio recorders and trackers (magnetic, up close) · Bluetooth trackers · cameras on the Wi-Fi
- **Your phone:** stalkerware, tracker SDKs, apps using the camera or mic right now
- **Your safety:** SOS (power x3, even locked) · fake call · siren · share location · 112 / 181 / 1930
- **Proof:** photo + SHA-256 fingerprint · complaint in English / Hindi (BNS s.77, IT Act s.66E)

Show this page (or the Detectors screen) and say one line.

**SAY**

Everything on this page is **in one app**, it works **offline**, and it speaks **English, Hindi and Telugu**.

### ● Find cameras and hidden devices

- **Full room scan:** the agent runs every detector and walks you round the room
- **Camera shape AI:** CCTV, dome, bullet cameras; phones used as cameras; laptop webcams
- **Lens finder:** the flash glint of a hidden lens
- **Night vision:** infrared LEDs of night cameras
- **Magnetic detector:** cameras, audio recorders, trackers and other powered electronics, up close
- **Mirror test:** two-way mirrors
- **“In view”:** names hiding spots (clock, frame, socket...) and ticks them off
- **Recorded video** analysis, with the hackathon hall as a built-in reference video

### ● Radio and network

- **Wi-Fi scan:** camera hotspots and camera makers' radios
- **Bluetooth scan:** AirTag, Tile, SmartTag trackers and Bluetooth cameras
- **Network scan:** cameras on the Wi-Fi you joined (RTSP / ONVIF / DVR ports)
- **Network check:** is this Wi-Fi private? Fake hotspots, open Wi-Fi, proxy, who sees your browsing
- **Your own phone**
  - **Phone check:** 158 known stalkerware / monitoring apps, hidden apps, one-tap uninstall
  - **App trackers:** ~60 tracking SDKs inside your apps
  - **Camera and mic in use** right now; **alerts** if they are used while the screen is off

### ● Personal safety

- **SOS:** press the power button 3 times (even locked), shake, or the Quick Settings tile → countdown → SMS with your location to your trusted contact · Call 112
- **Fake call (secret call):** your phone rings with a real-looking call screen, now or after 10 s / 30 s / 1 min, even with the phone locked
- **Siren:** loud alarm with the flash blinking
- **Share my location:** a map link plus what TwinGaze found
- **Helplines:** 112 · 181 women · 1930 cyber crime
- **Protection outside the app:** keeps working when TwinGaze is closed

### ● Evidence, report, ease of use

- **Evidence:** time-stamped photo + SHA-256 fingerprint · evidence locker · history
- **Report:** agent verdict + complaint in English / Hindi (BNS s.77, IT Act s.66E, 1930)
- **Send report to laptop** with vivo Office Kit
- **Voice guide:** English · हिंदी · తెలుగు · Off / Alerts / Guide · choose the voice
- **Discreet mode:** dim, silent, guidance by vibration
- **Sensitivity:** Sensitive / Balanced / Strict
- **No account, no cloud:** works in airplane mode

One codebase, native where it matters, AI on the phone.

**SAY**

Here is how we built it. The app is written in plain **HTML, CSS and JavaScript**, with no heavy framework, so the **same code** runs as an Android app and as a website.

We wrap it with **Capacitor** for Android, and we wrote **four native Java plugins, a background service and a Quick Settings tile** for what a web page cannot do: the magnetometer, Wi-Fi and Bluetooth scanning, the list of installed apps, the voice, and the SOS.

The AI runs on the phone through **ONNX Runtime**, on the phone's **GPU using WebGPU**, with a CPU fallback, in a background worker, so the camera never freezes. We trained our model with **PyTorch and Ultralytics YOLO** on a laptop GPU. And with **vivo Office Kit**, the report goes from the phone to the laptop in one tap.

| LAYER                         | WHAT WE USED                                                                                                                                                                        |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>App (UI + logic)</b>       | HTML5 · CSS · JavaScript (~7,000 lines) · one codebase for the Android app and the website · offline PWA (service worker)                                                           |
| <b>Android</b>                | Capacitor 8 · Java (~3,400 lines) · 4 native plugins (sensors & voice, radio, phone check, protection) + a foreground service + a Quick Settings tile · Android 7 to 16 (SDK 24–36) |
| <b>On-device AI</b>           | ONNX Runtime Web 1.30 · WebGPU (GPU), WebAssembly SIMD (CPU) fallback · runs in a Web Worker                                                                                        |
| <b>Model training</b>         | Python · PyTorch · Ultralytics YOLO11 · NVIDIA RTX 5050 laptop GPU · 100 epochs                                                                                                     |
| <b>Auto-labelling</b>         | Google OWLv2 + Grounding DINO (two open-vocabulary detectors must agree on every box)                                                                                               |
| <b>Phone hardware</b>         | Camera + flash · magnetometer · gyroscope / orientation · accelerometer · light sensor · Wi-Fi · Bluetooth LE · GPS · vibration · speaker                                           |
| <b>Voice &amp; sound</b>      | Android text-to-speech (offline Indian voices, navigation audio channel) · Web Audio beeper                                                                                         |
| <b>Evidence &amp; privacy</b> | IndexedDB (stays on the phone) · SHA-256 fingerprints (Web Crypto) · no accounts, no cloud                                                                                          |
| <b>Phone ↔ laptop</b>         | vivo Office Kit: report via share sheet, evidence album sync, shared clipboard, screen mirroring                                                                                    |

**KEY POINTS**

- **Same code** = app + website · **native Java** only where the web can't reach · **AI on the phone's GPU**, CPU fallback, in a worker

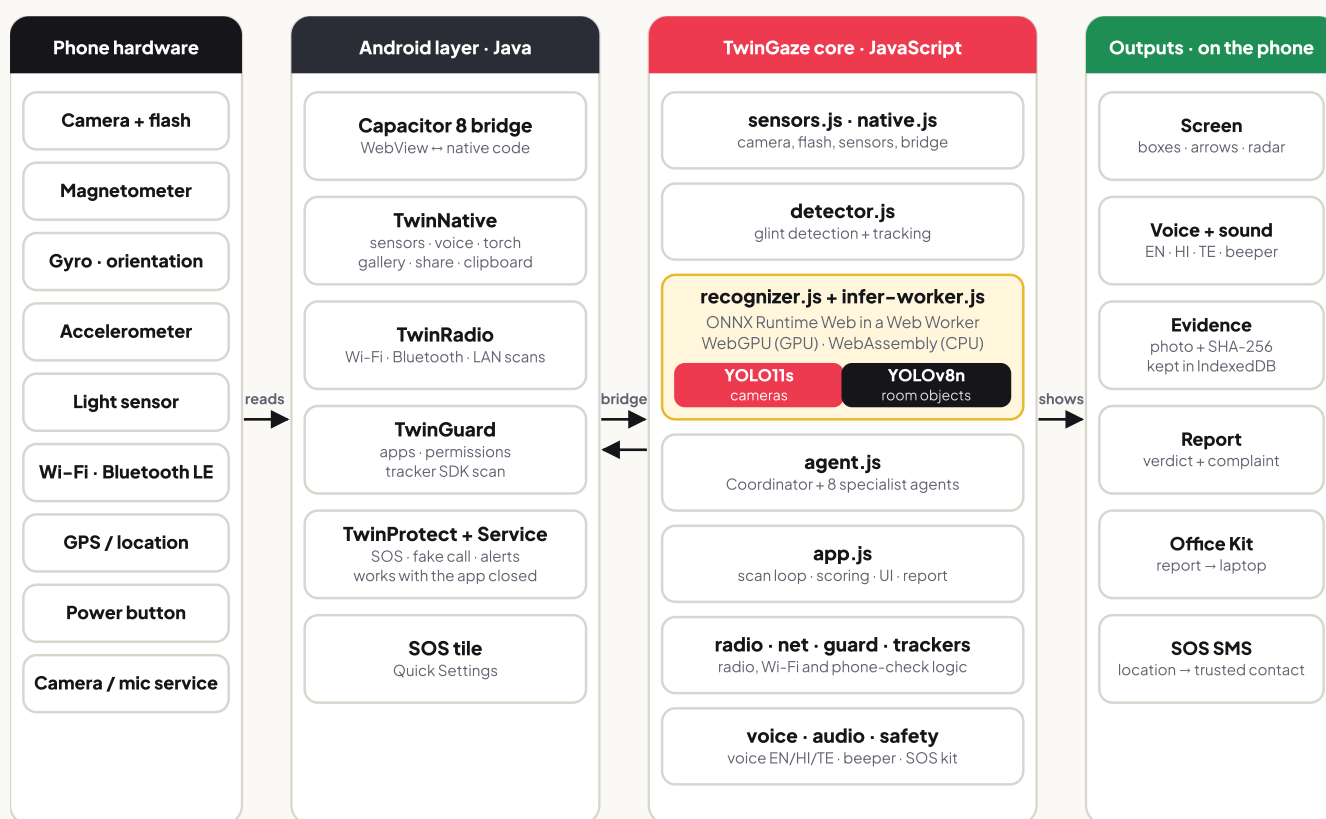
## 4b Architecture

~30 s

Show the diagram and walk left to right.

### SAY

On the left is the phone's own **hardware**. Our **Android layer**, written in Java, reads it and hands it to our **JavaScript core** through Capacitor. The core runs the glint detector, the **two AI models on the GPU**, and the agent. The results come out as boxes, voice, evidence and the report, and **all of it stays on the phone**.



Everything runs on the phone · no cloud · no account · nothing is uploaded · works in airplane mode

The same JavaScript core also runs as the website (offline PWA): browser APIs take the place of the Android layer.

### KEY POINTS

- **4 layers:** hardware → Android (Java) → core (JavaScript + ONNX Runtime) → outputs
- **No server:** nothing leaves the phone except what you choose to send (SOS SMS, report)

Two neural networks on the phone. Two big models only to label the training data. No LLM.

## SAY

We use **two neural networks**, both running on the phone.

The first is **our own camera detector: YOLO11s**, 9.4 million parameters, fine-tuned for TwinGaze. We collected about **1,100 freely licensed photos** of CCTV cameras, rooms and ceilings. To label them we used two large AI models, **OWLv2 and Grounding DINO**, and kept a box only when both agreed. We added smoke detectors and phone look-alikes, so the model learns what is *not* a camera, and the **real false alarms from this hall**. On photos it had never seen, it found **12 of 18 cameras with zero false alarms**.

The second is **YOLOv8n** from Open Images. It names **49 kinds of room objects**, like clocks, frames, sockets and smoke detectors, so the agent can say “Clock ahead, check it”.

There is **no chatbot and no LLM** inside. The agent is a set of rules. That makes it **instant, predictable and fully offline**.

## MODEL 1 • TRAINED BY US

## Camera detector: YOLO11s

9.4 M parameters · 36 MB ONNX · 416×416 input

Classes: **camera** **smoke detector** **phone**

Data: ~1,100 Wikimedia Commons photos + 306 phone photos + 238 look-alikes (remotes, power banks...) + hall false alarms as hard negatives

Held-out test: **12/18 cameras, 0 false alarms** in

33 camera-free photos (old YOLO11n: 2 false)

## MODEL 2 • PRE-TRAINED

## Room objects: YOLOv8n

Open Images V7 (Ultralytics) · 13.5 MB · 416×416 input

49 classes kept: person, clock, TV, picture frame, mirror, lamp, socket, fan, soft toy, plant, smoke detector...

Used for: hiding-spot hints, the “In view” list, the room checklist, phones and laptops near a glint

## TRAINING ONLY (NOT IN THE APP)

OWLv2 (Google) + Grounding DINO: auto-labelling

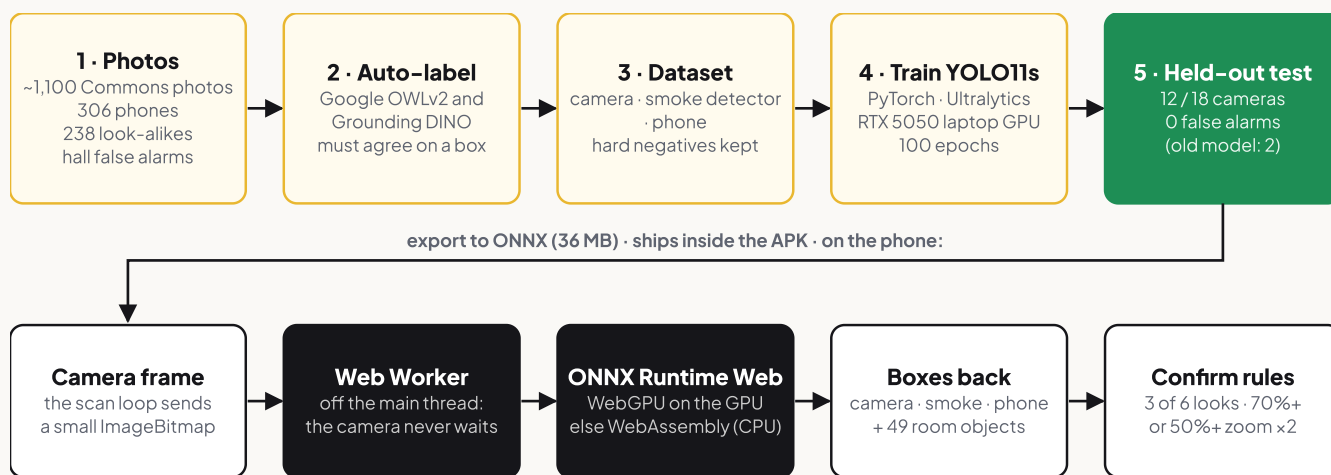
## KEY POINTS

- **A camera box counts only after 3 of the last 6 looks**, at 70%+, or 50%+ when the zoomed look agrees twice
- **No LLM**: rule-based agent → instant, deterministic, private, works in airplane mode

Top row: training on the laptop. Bottom row: what happens on the phone.

## SAY

We labelled about 1,100 photos automatically, with **two big AI models that had to agree**, added the **false alarms from this hall**, and trained **YOLO11s** on a laptop GPU. On the phone it runs in a **background worker** on the GPU, so the camera never freezes.



Speed: the scan measures how long a check takes and spaces the checks to match (up to 5 checks a second on the GPU).

Room objects: YOLOv8n (Open Images V7, pre-trained), 49 classes kept, same runtime.

| HELD-OUT PHOTOS (NEVER SEEN IN TRAINING)               | CAMERAS FOUND (OF 18) | FALSE ALARMS (33 CAMERA-FREE) |
|--------------------------------------------------------|-----------------------|-------------------------------|
| Off-the-shelf zero-shot detectors                      | 4                     | 5                             |
| YOLO11n (2.6 M) · our first model, at 50%              | 12                    | 2                             |
| YOLO11n at 70%                                         | 7                     | 1                             |
| <b>YOLO11s (9.4 M) · in the app, at 50% and at 70%</b> | <b>12</b>             | <b>0</b>                      |

## KEY POINTS

- **Auto-labelling** (OWLv2 + Grounding DINO): no hand labelling, and two models must agree
- **Hard negatives** from the real hall taught it what is *not* a camera

What makes it trustworthy: no false alarms, small far cameras, real guidance.

#### SAY

A few technical points we are proud of. **First, no false alarms from shiny things.** A bright point is only called a camera after a chain of tests: it must go dark when we switch the flash off, it must stay on its object when you move sideways, and it must pass the flash test twice. Even then, a glint alone is not enough. We need a **second, independent sign**, like a camera shape or a magnetic jump. Otherwise it is shown as a “possible lens”, never as a camera.

**Second, small and far cameras.** The AI looks at the whole picture, then zooms into the middle at **two and four times** the detail. That is how it finds a CCTV dome across a hall.

**Third, the agent.** A coordinator runs the scan as a plan (prepare, radio, survey, inspect, report) with specialist agents that share one memory of the room. It uses the orientation sensor to say “Turn right 60 degrees, to the clock”.

**OPTIONAL** · And we tested it properly: 83 automated tests and full-scan simulations. The video of this hall is built into the app as a reference. TwinGaze finds both CCTV cameras in about two seconds, every time.

#### 1 · Flash test

goes dark when the flash is off? (lamps stay lit)

#### 2 · Sideways

6°+ of parallax from the orientation sensor

#### 3 · Stays put

moves with its object (block matching), not across a shiny surface

#### 4 · Flash again

still the same reflection after moving

#### 5 · 2nd sign

camera shape · magnetic +8  $\mu$ T · phone/laptop at the spot

#### Camera found

else “possible lens”

#### KEY POINTS

- **Agent:** Coordinator + 8 specialists (Environment, Radio, Coverage, Scene, Glint, Magnetic, Navigator, Report) · 12×30° coverage radar
- **Voice manager:** 4 priority levels, one sentence at a time, “Camera found” always cuts in
- **Protection outside the app:** Android foreground service: power x3 SOS, alerts when the camera/mic is used with the screen off
- **Evidence:** SHA-256 per photo, stored only on the phone, complaint in English / Hindi
- **Proof:** 83 automated tests · hall reference video: 2 of 2 cameras in 5 of 5 runs, no false cameras · busy hall: instruction changes 56 → 20 per minute

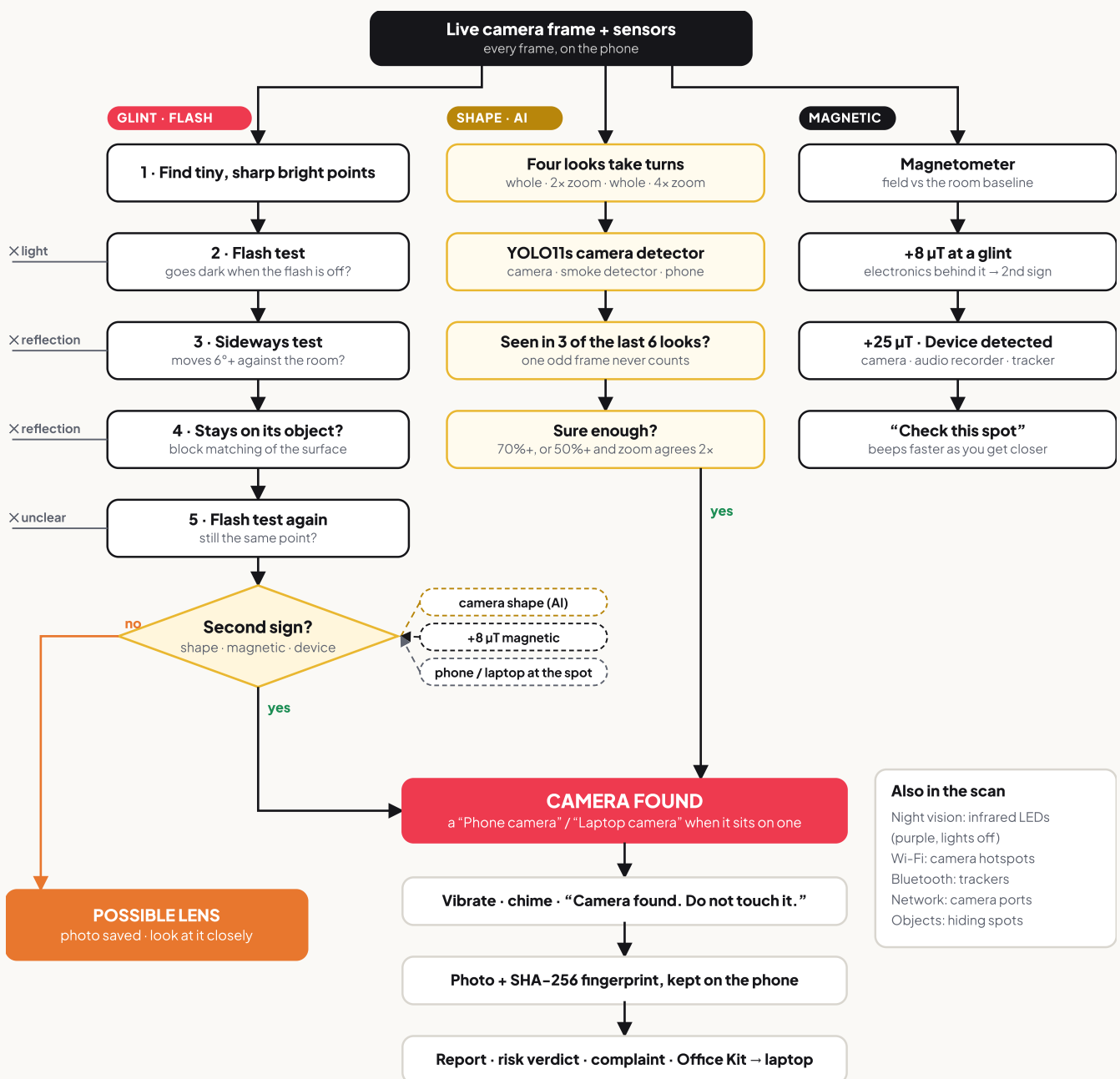
# Flow chart: how a camera is confirmed

~40 s

Three lanes run on every frame. A glint needs a second sign.

## SAY

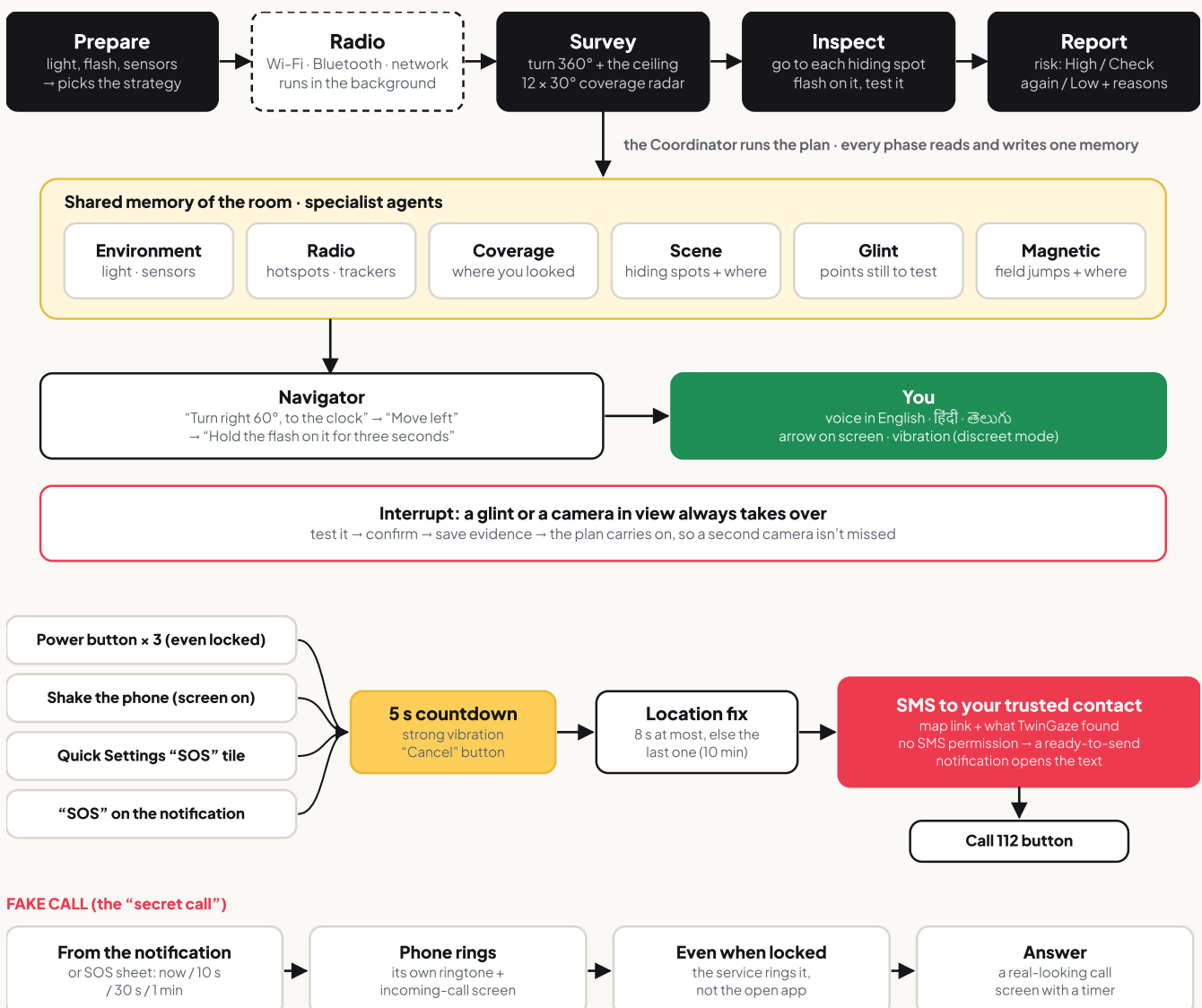
Three lanes run on every frame. The **glint** lane rules out lights and reflections step by step, and then needs a **second sign**. The **AI** lane confirms a camera by its shape. The **magnetic** lane finds hidden electronics, like an **audio recorder**, up close.



How the full room scan is run, and what happens in an emergency.

### SAY

The **agent** runs the scan as a plan and remembers the room, so it can say “Turn right 60 degrees, to the clock”. And in an emergency, **three presses of the power button** send your location, even with the phone locked.



Cue sheet: do, then say one line. Mirror the phone on the laptop with Office Kit first.

## DO → SAY

1. **Home** → **“Watch it find cameras · reference video”** (one tap, under Scan room).  
Both ceiling domes get a red CAMERA box.  
*“This is a video of this hall. It finds both CCTV cameras in about two seconds.”* The report shows **Reference check: PASS, 2 of 2.**
2. **Live: point the phone at the hall's CCTV dome.** Chime + “Camera found”.  
*“Lights on, shape recognition. No internet needed.”*
3. **Lens finder on another phone's back camera** (a phone left on a shelf records like a spy camera).  
*“The flash bounces off the lens; the phone is recognised at the same spot: phone camera found.”*
4. **Magnetic: bring the phone to a charger or speaker.** The beeping speeds up.  
*“Magnetic alone never calls it a camera; it only adds weight.”*
5. **Finish** → **report** → **“Send report to laptop · Office Kit”**. Open it on the laptop.  
*“Photo, fingerprint, and a complaint ready for 1930, in English or Hindi.”*
6. **SOS button (top right) → Fake call → Now.** The phone rings with a real-looking call screen.  
*“A fake call gives you a reason to leave. And even with the app closed, pressing the power button three times sends an SOS with your location.”*

## SAY · CLOSING

Ananya does not need to buy anything. With TwinGaze, the phone she already owns checks her room in two minutes, guides her in her own language, keeps the evidence, and helps her get safe. **It runs on the phone, it respects her privacy, and it speaks her language. Thank you.**

Before the demo: charge both phones, turn Voice to “Guide”, set the language, and keep the room lights on for step 2. The siren is loud: only show the SOS sheet, don't press Siren.

Say the limits before a judge finds them. It builds trust.

## What it can't do

- It can miss a camera that is covered, pointed away, or behind tinted plastic. "No camera found" lowers the risk; it doesn't rule it out.
- A pinhole camera in a clock looks like a clock to the shape AI. That is what the glint and magnetic tests are for.
- A CCTV dome far across a big hall is at the edge of the AI: walk closer and point at it.
- A tiny shiny bead or screw head can pass the light tests, so it stays a "possible lens" until a second sign confirms it.
- Smoke detectors are sometimes confused with dome cameras.
- Radio scans need the Android app, and many cameras record to a memory card with no Wi-Fi at all.

## Likely questions

### Does it need internet?

No. Models, voice and everything else run on the phone. Nothing is uploaded.

### Why not an LLM?

It must be instant, offline and predictable. Rules are deterministic and fast; an LLM on a phone is slow and can make things up.

### How accurate is it?

12 of 18 cameras with 0 false alarms on unseen photos; 2 of 2 domes in the hall video, 5 of 5 runs.

### How do you avoid false alarms?

Two independent signs are needed; otherwise "possible lens". We trained on real false alarms from this hall.

### Which phones?

Android 7 to 16. GPU (WebGPU) when the phone has it, CPU otherwise, just slower.

### How is it different from other apps?

Other apps each do one thing (often only the magnetic sensor). TwinGaze puts them in one app, and the signals confirm each other; plus guidance, evidence, safety and Indian languages, all offline.

### Can it find hidden audio recorders?

Up close, yes: the magnetic detector reacts to any powered electronics (recorder, tracker, camera). It can't tell which device it is, so it says "Electronic device detected, check this spot".

## IF YOU REMEMBER ONLY THIS

- **Problem:** hidden cameras in PGs, hotels and trial rooms; people can't check
- **Solution:** TwinGaze: shape AI + glint + magnetic + night vision, on the phone, offline
- **Proof:** our own YOLO11s: 12/18 cameras, 0 false alarms; hall video: 2 of 2
- **Impact:** evidence + complaint + SOS, in English, Hindi and Telugu